

B.Tech. (Bio-Tech) 3rd Semester G-Scheme
Examination, December-2025
BIOLOGY
Paper-BSC-BIO-201-G

Time allowed: 3 hours

Maximum marks: 75

Note: Attempt any five questions, selecting one question from each unit. All questions carry equal marks.

1. Write short note on the following: **6x2.5=15**
- a) Mendel's Second law of inheritance
 - b) Watson and Crick model of DNA
 - c) Plasmid Vectors
 - d) Sewage treatment
 - e) Differentiate between glucose and cellulose.
 - f) Genetically modified crops

Unit-I

2. Distinguish between Prokaryotic and Eukaryotic cells. Explain the key differences in their structure and function. **15**

OR

3. Describe the central dogma of molecular biology and explain the concept of genetic code. How does it relate to the synthesis of proteins? **15**

Unit-II

4. Explain the structure and functions of proteins, including the primary, secondary, tertiary and quaternary levels of organization. Provide examples of proteins with different functions. **15**

OR

5. Discuss the structure and functions of lipids, specifically phospholipids and cholesterol. How do they contribute to the structure and function of cell membranes? **15**

Unit-III

6. Describe the role of restriction enzymes and DNA modifying enzymes in recombinant DNA technology. How do they facilitate gene cloning? **15**

OR

7. What are transgenic plants and animals? Discuss their applications and potential benefits in medicine. **15**